Systematics Violations of Error Assumptions

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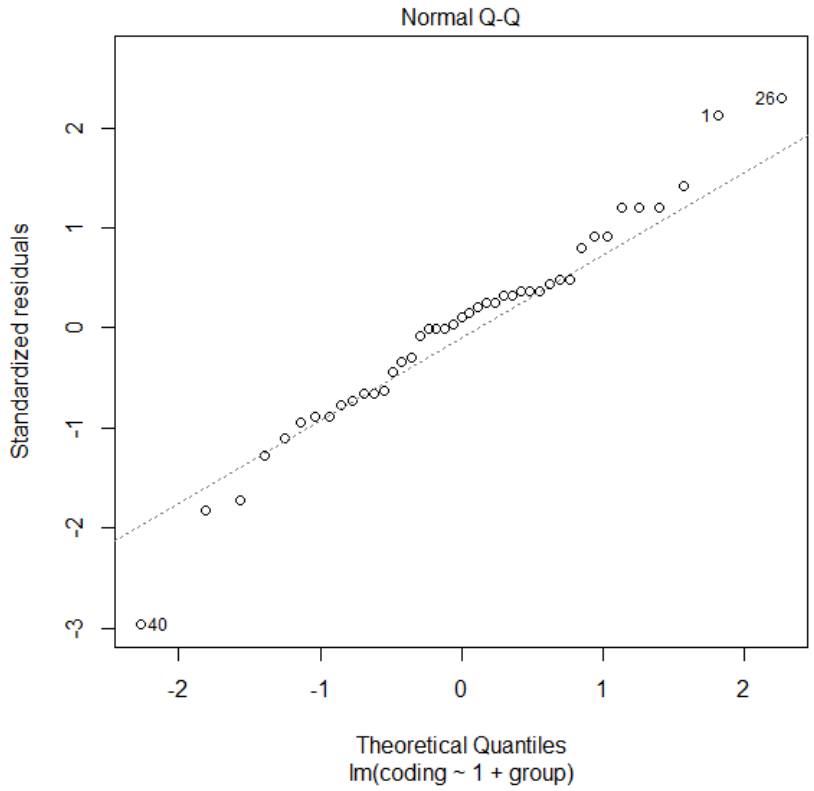
1. Normality

- ***Assumption***: Our residuals follow a normal distribution.

- This assumption can be checked with a quantile-quantile normal plot, histogram, or density plot of the residuals.

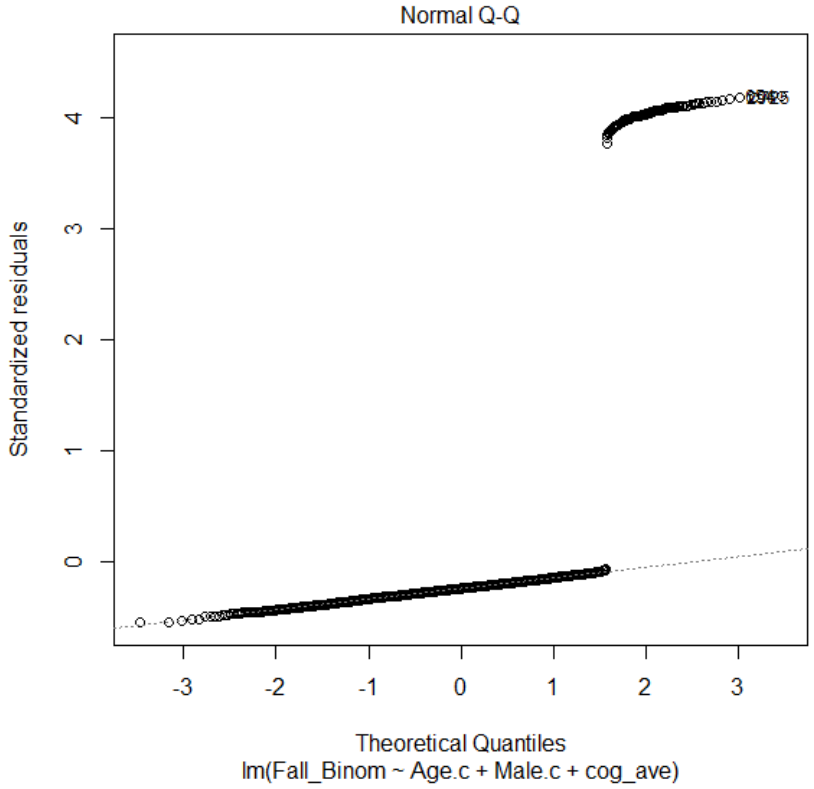
- It can also be checked with goodness of fit tests like the ***Shapiro-Wilk*** test or the ***Kolmogorov-Smirnov*** test.
  - But remember that violation is continuous not categorical.



1. Normality

- ***Assumption***: Our residuals follow a normal distribution.

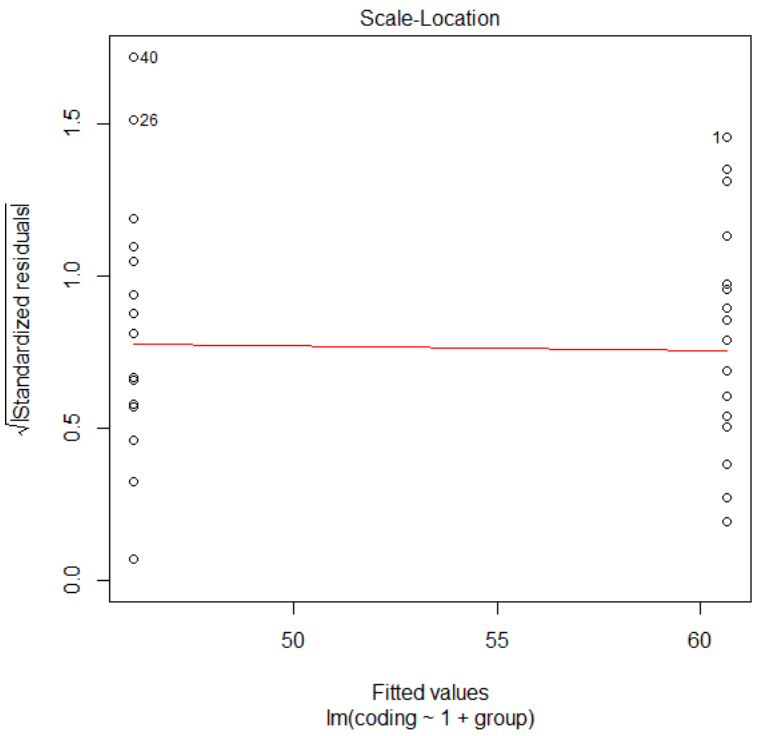
- **What happens if we violate this assumption?**
  - Violation of the normality assumption does not create bias or inefficiency in our estimation.
  - However, it does affect our error terms and thus the calculation of p-values.
  - ***The more your residuals depart from normality, the less reliable your p-values are.***



2. Homoscedasticity

- ***Assumption***: Our residuals have an equal variance across all levels of the fitted-values.

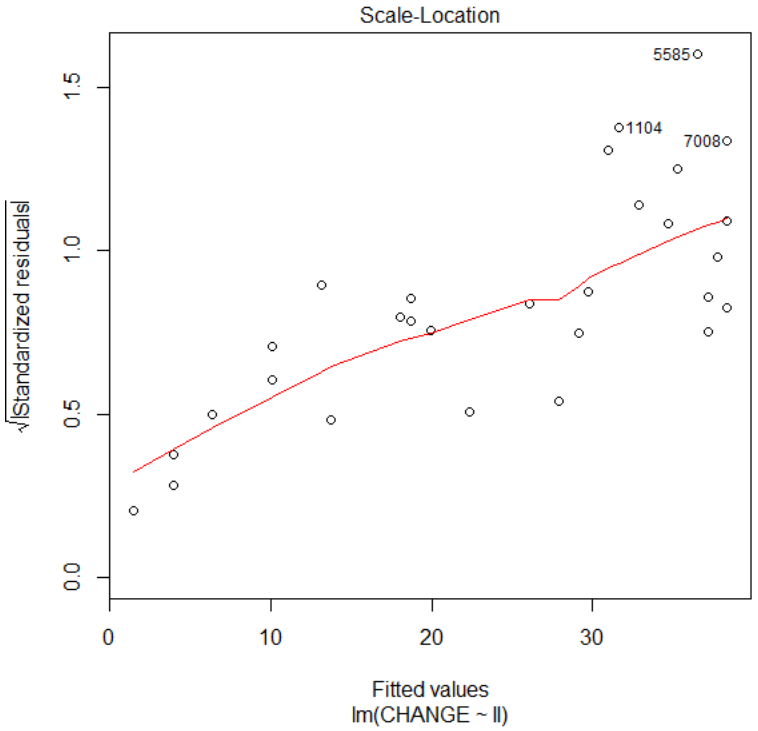
- This assumption can be checked with a Scale-Location plot, or by regressing the standardized/studentized residuals onto the fitted values.



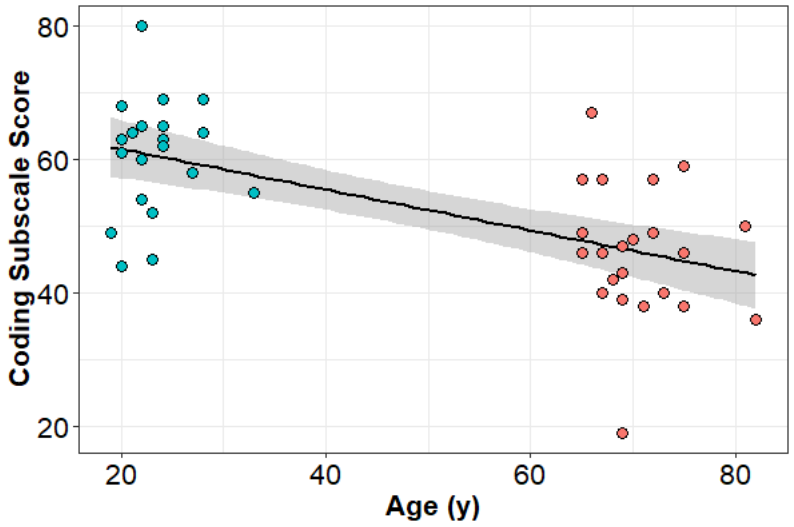
2. Homoscedasticity

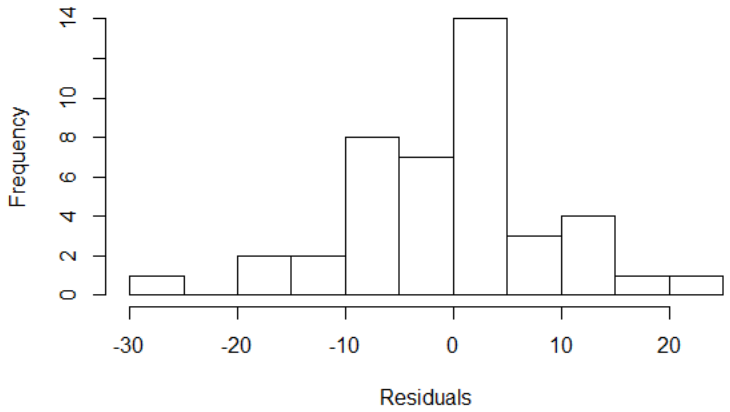
- ***Assumption***: Our residuals have an equal variance across all levels of the fitted-values.

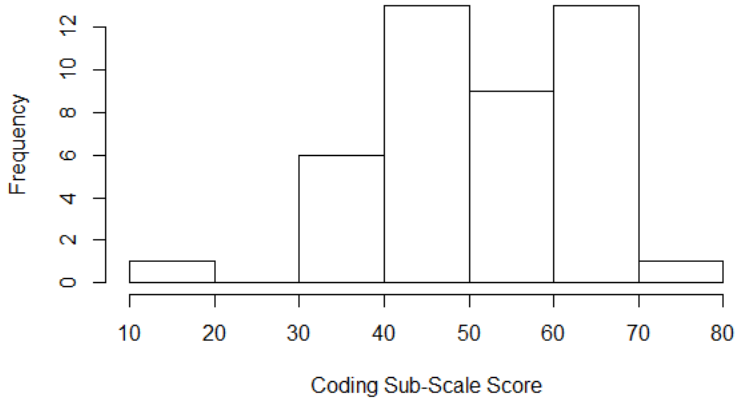
- **What happens if we violate this assumption?**
  - In OLS regression, we assume constant variance in the calculation of standard errors, which are necessary for the calculation of p-values.
  - ***The more heterogeneous your residuals, the less reliable your p-values are.***



3. Well-Behaved Residuals







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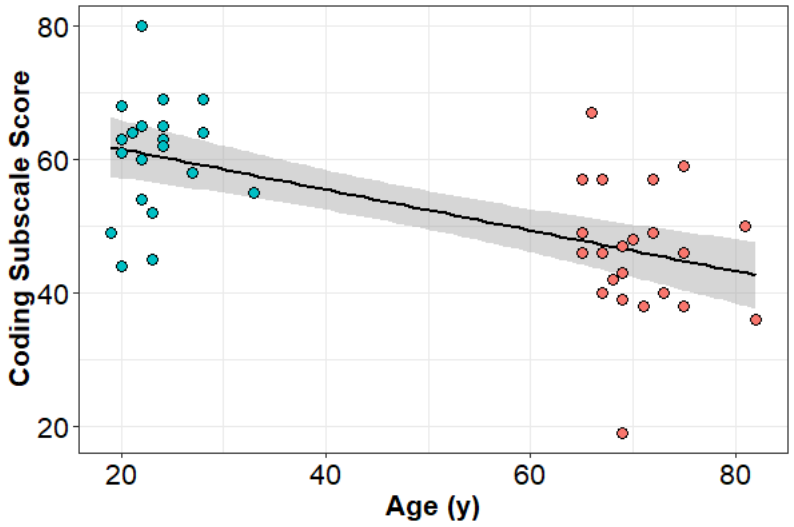

$$Coding \ = \beta_0 + \beta_1(Age\ Group_i) + \epsilon_i$$

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plot(model_name)
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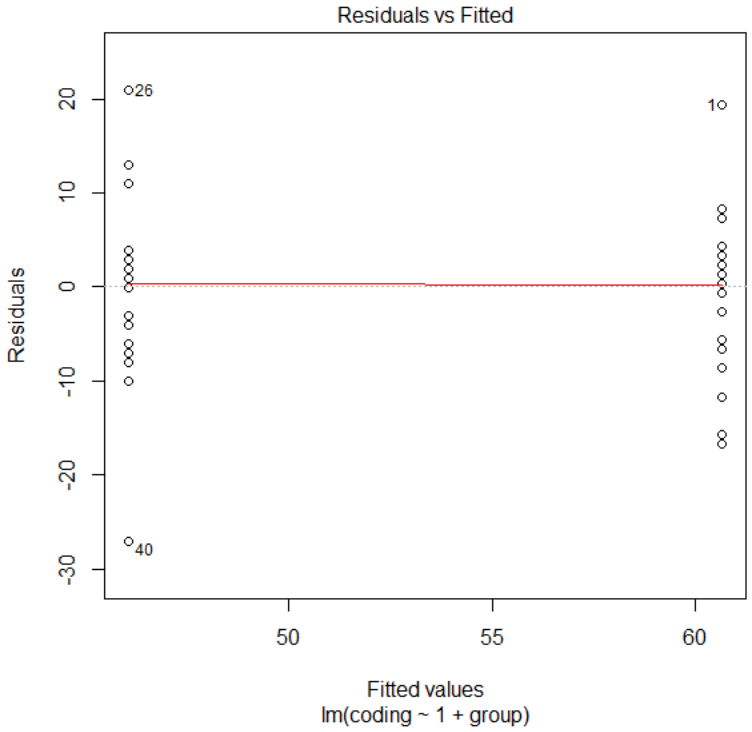

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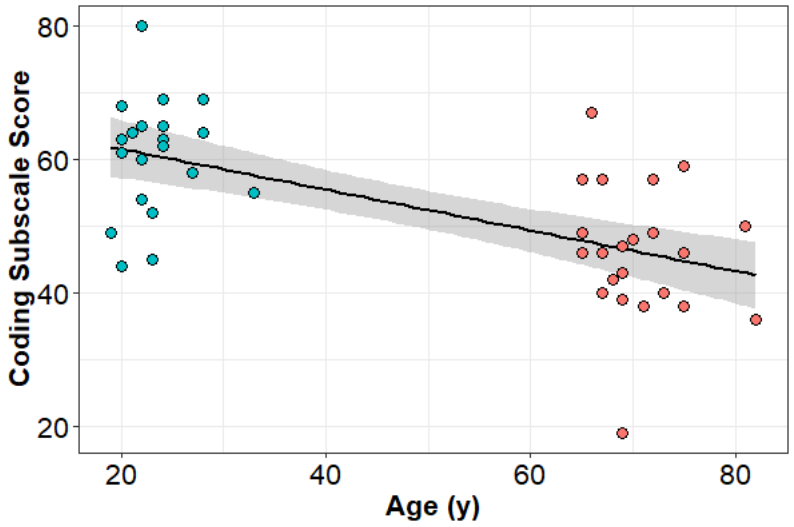

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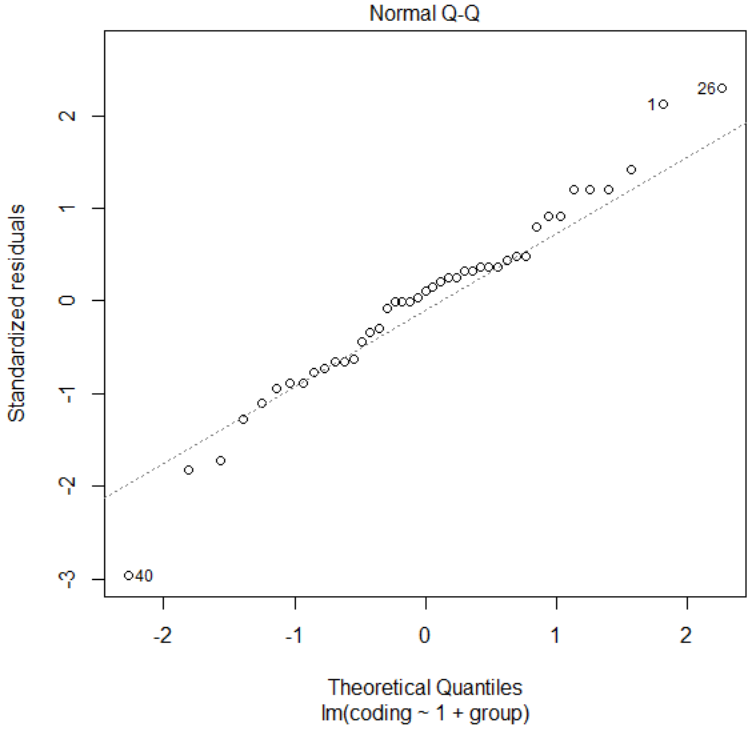

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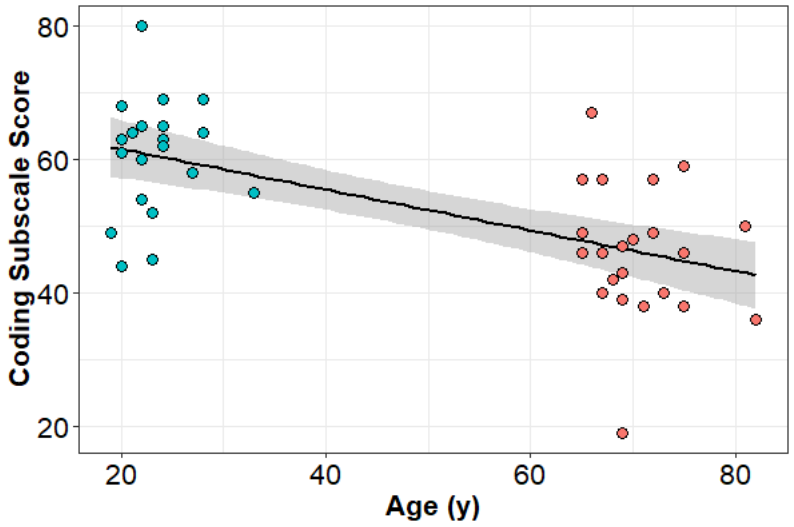

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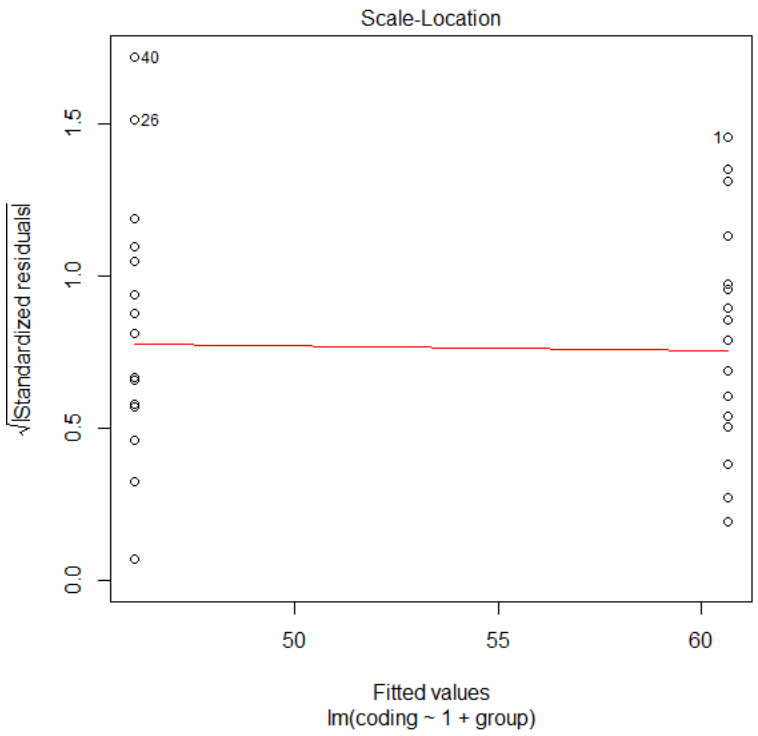

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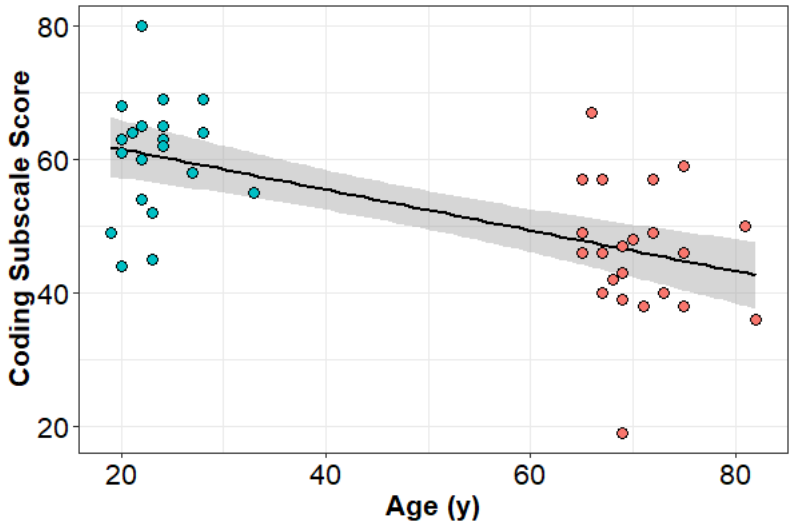

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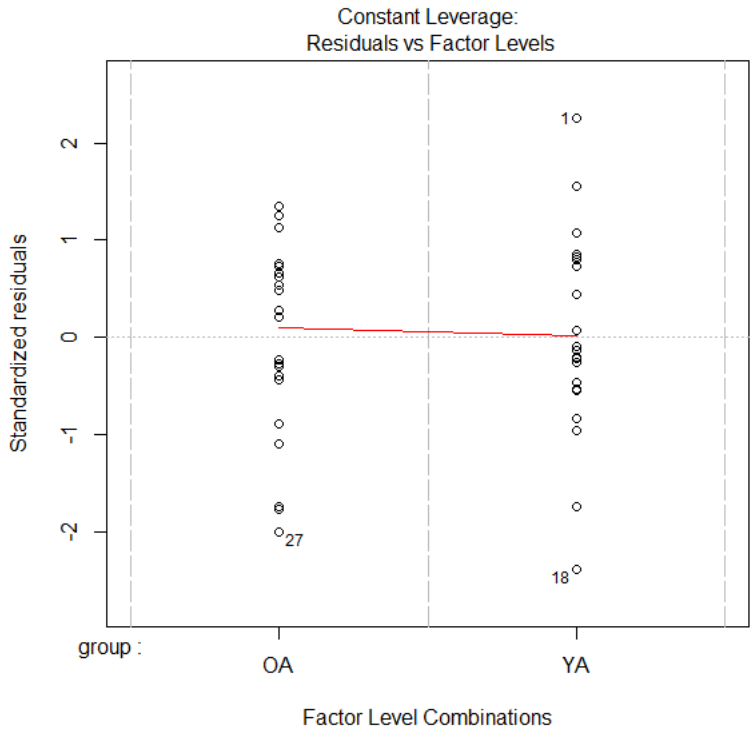
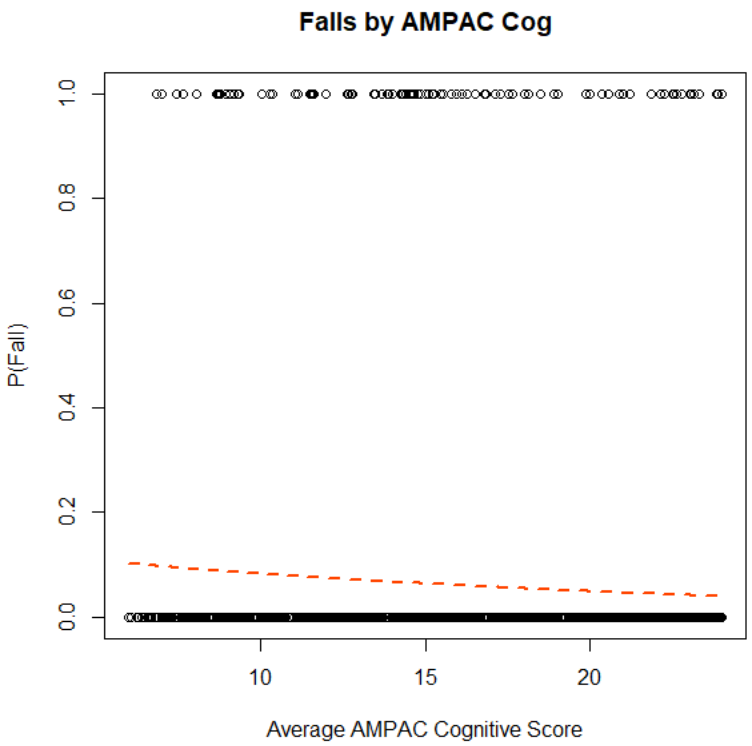


4. Clear Violations



In this example, we will be looking at a binomial outcome (did a person experience a fall or not) as a function of their score on a cognitive assessment.

We would never want to do this, as the binomial data will lead to substantial violations of our assumptions, but it is illustrative for educational purposes.

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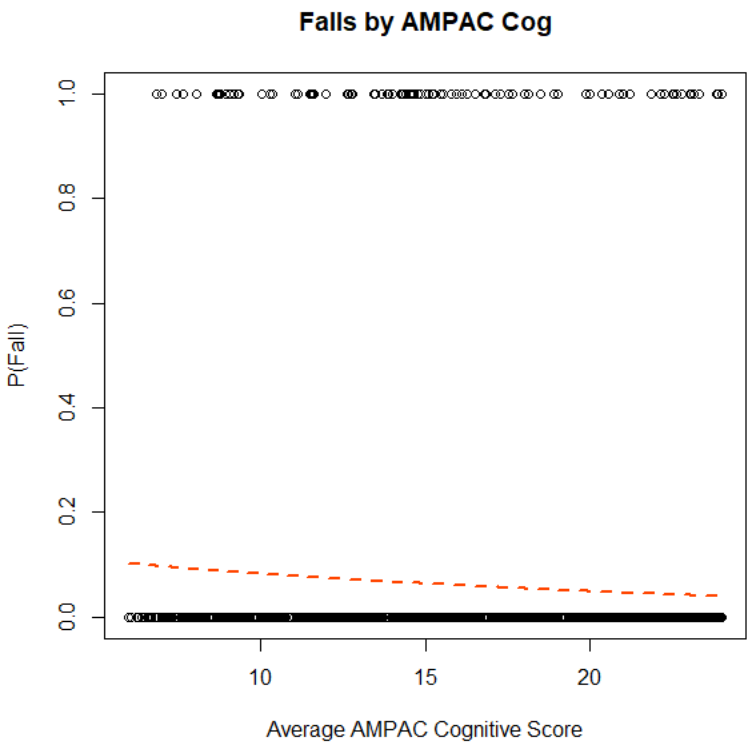

$$Fall = \beta_0 + \beta_1(AMPAC_i) + \epsilon_i$$

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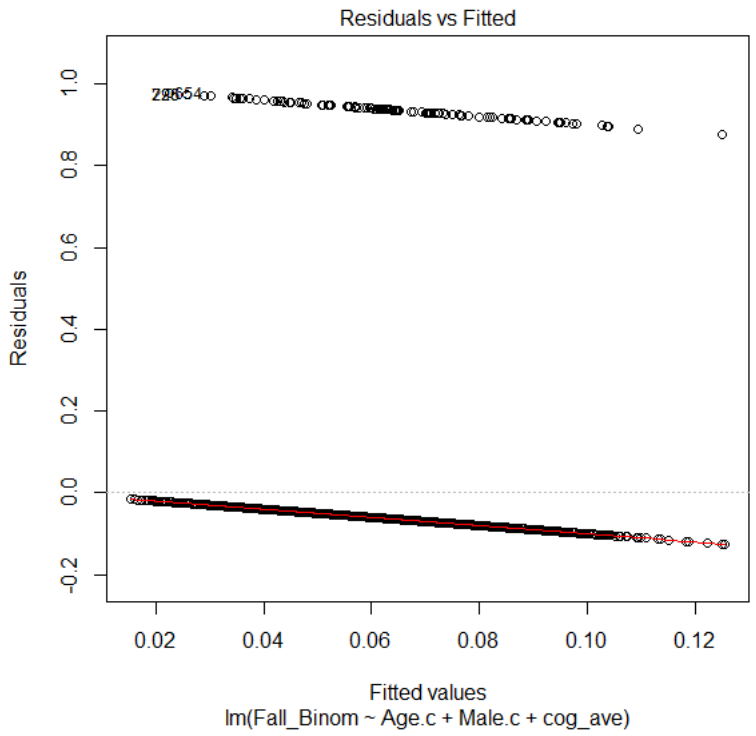

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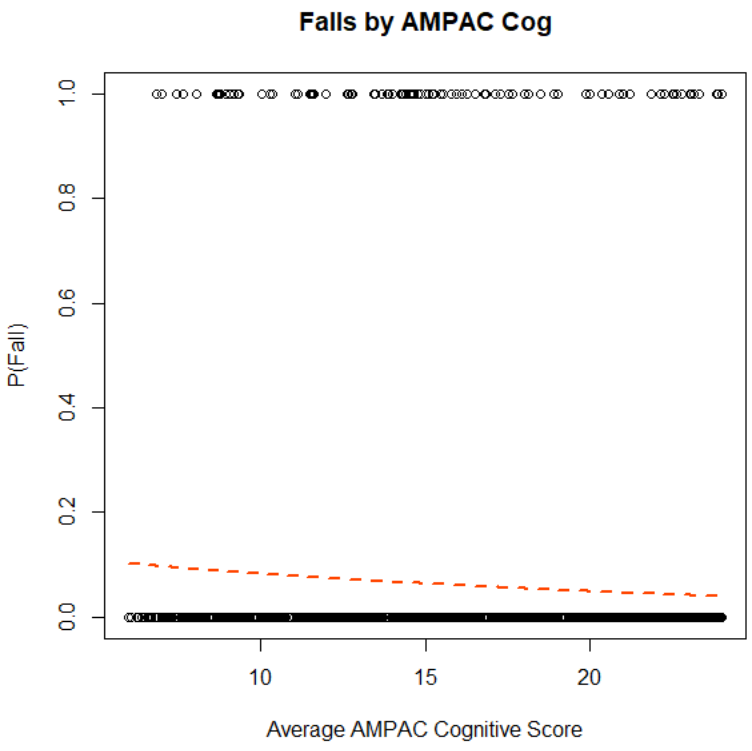

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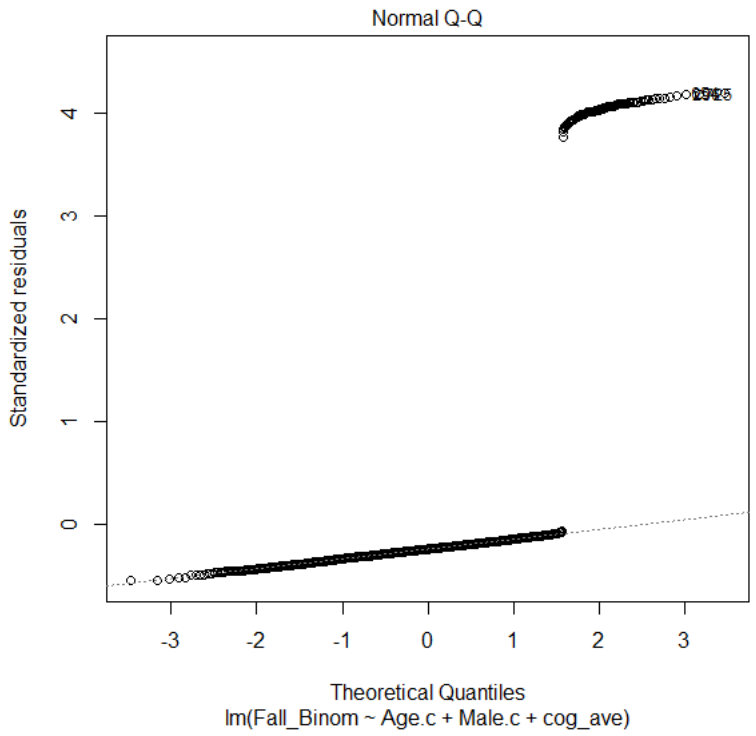

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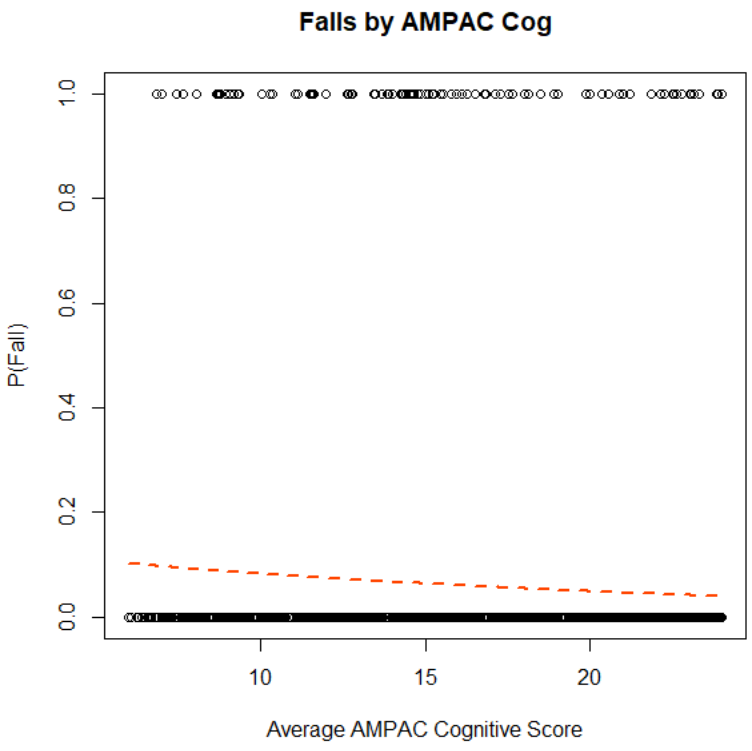

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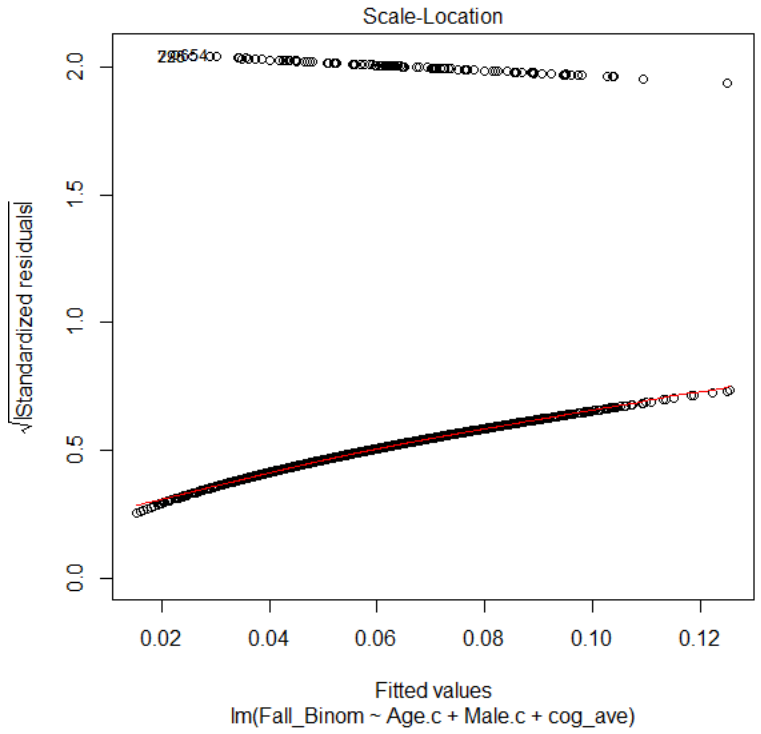

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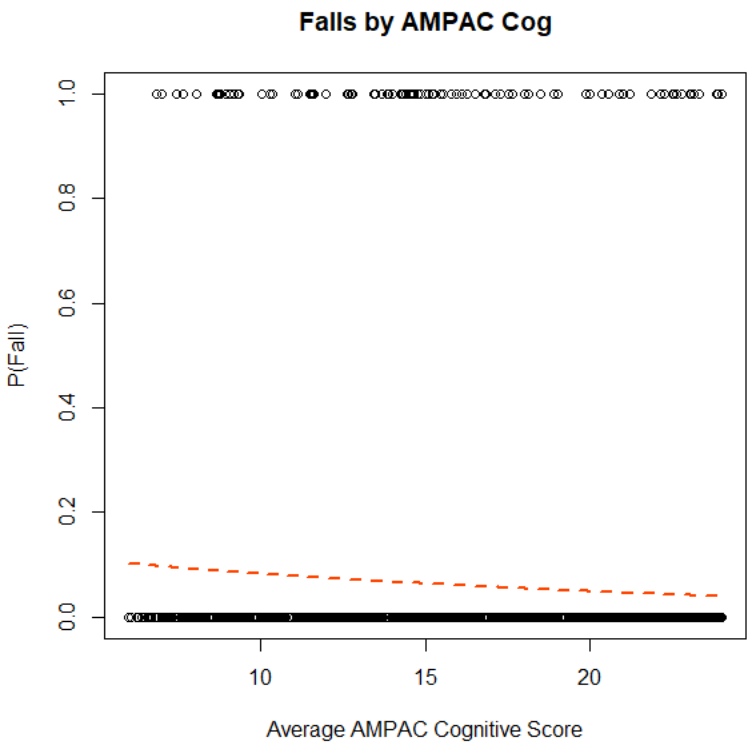

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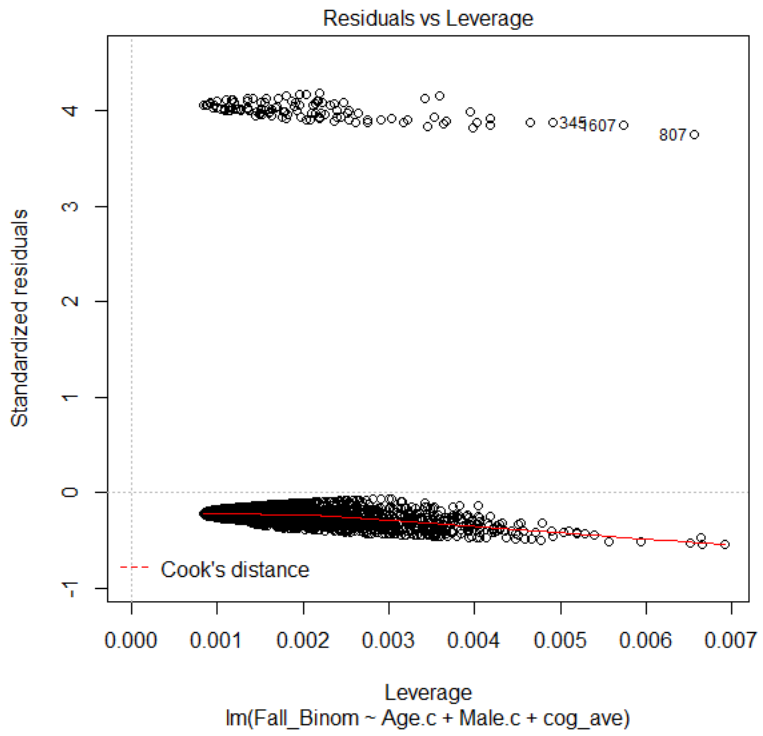
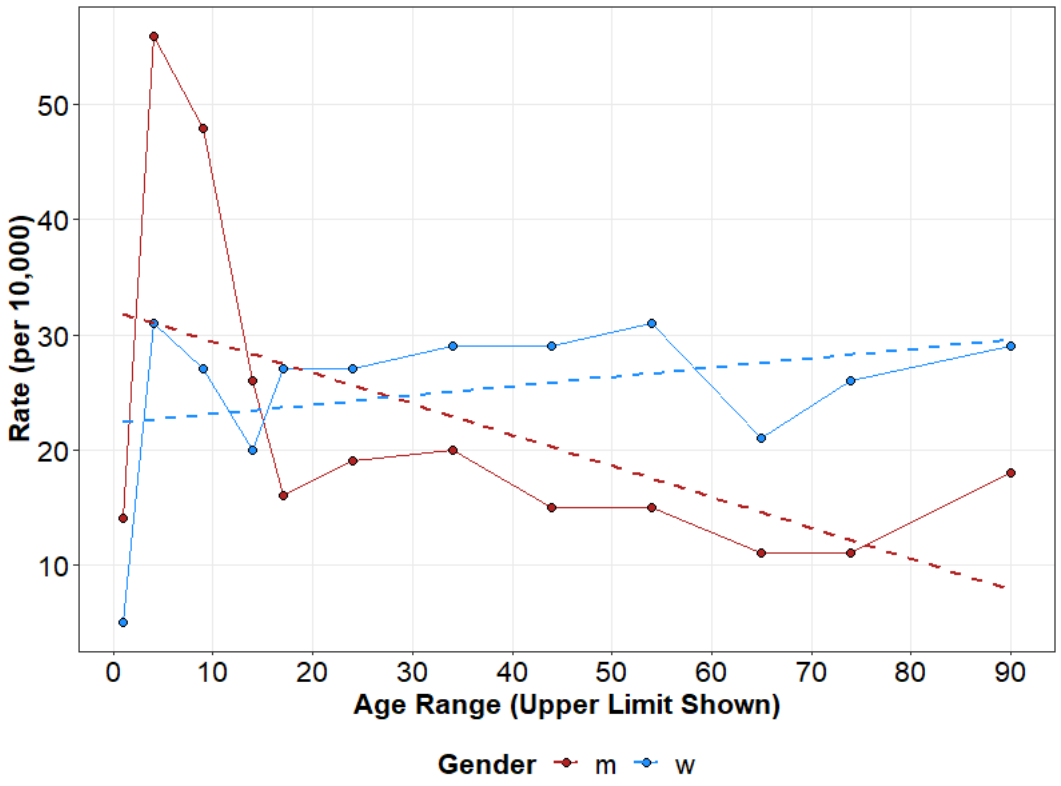
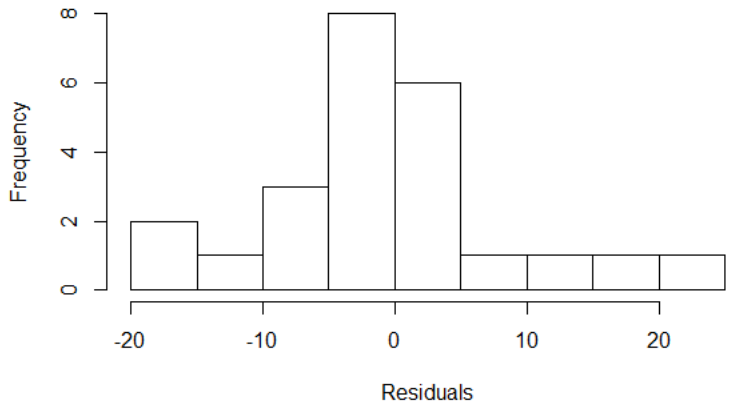


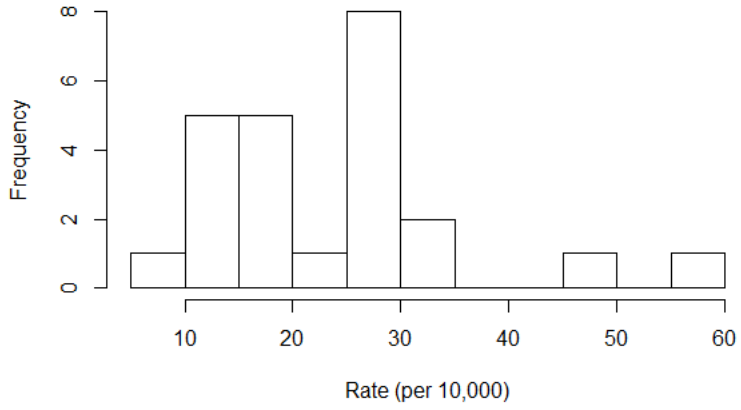
4. Clear Violations

- What do we do in these situations?
  - Generally, when our assumptions are this substantially violated, that means we have chosen a poor statistical model to explain our data.
  - This will require us to use “generalized linear models”, of which there are several families:
    - For instance, with a binomial outcome we should use a **logistic regression**.
    - For count data, we will need to use **poisson regression**.
    - For categorical data (e.g., choices among multiple candidate), we will need to use **multinomial logistic regression**.
    - For ordinal data (e.g., ranks), we will need to use **ordered logistic regression**.
  - Fortunately, everything we have learned carriers over, so you should be well suited to tackle these advanced topics in future classes.

5. Difficult Residuals







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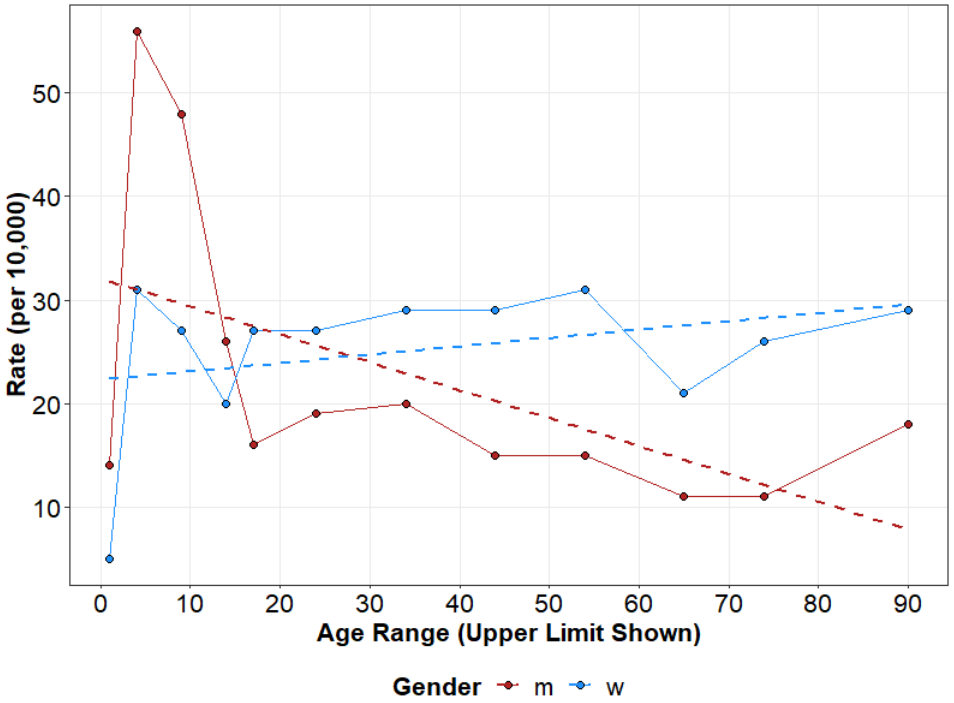

$$Rate \ = \beta_0 + \beta_1(A_i) + \beta_2(G_i) + \beta_3(AxG_i) + \epsilon_i$$

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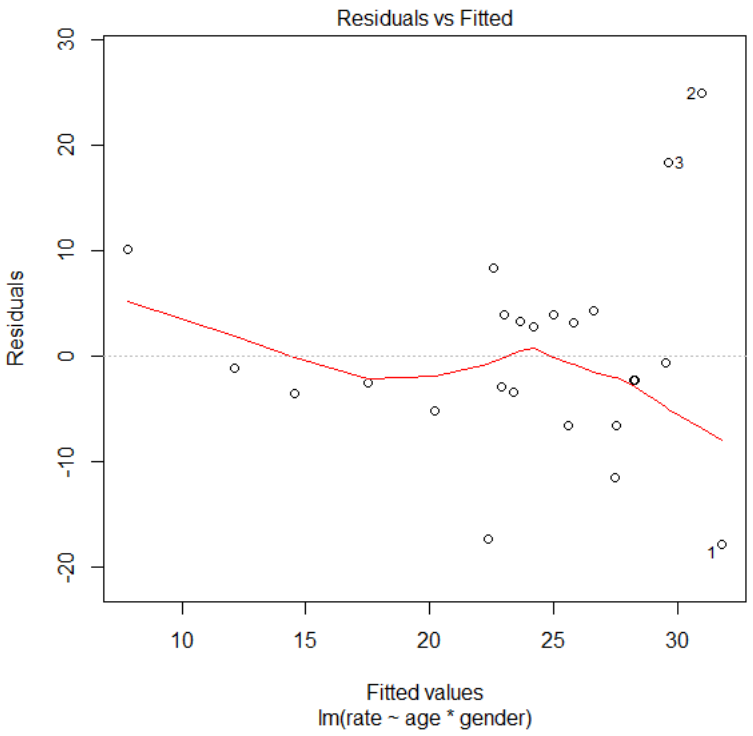

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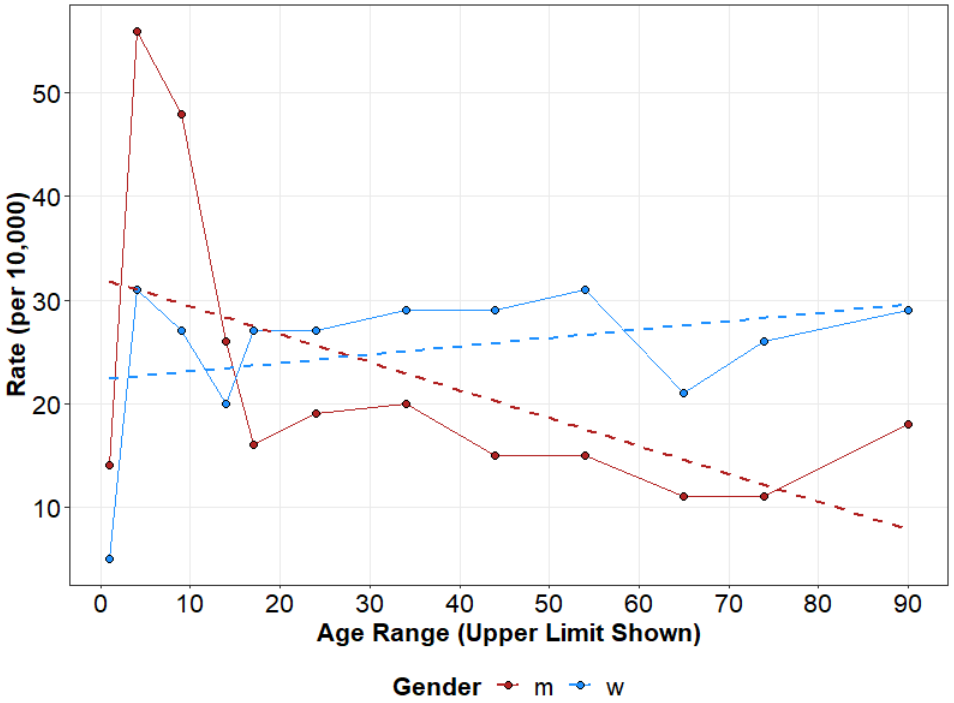

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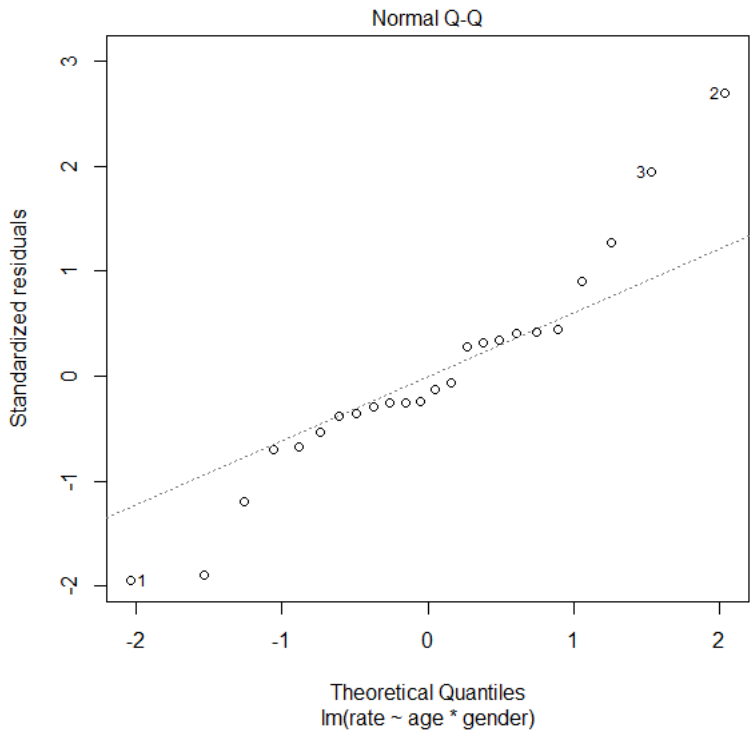

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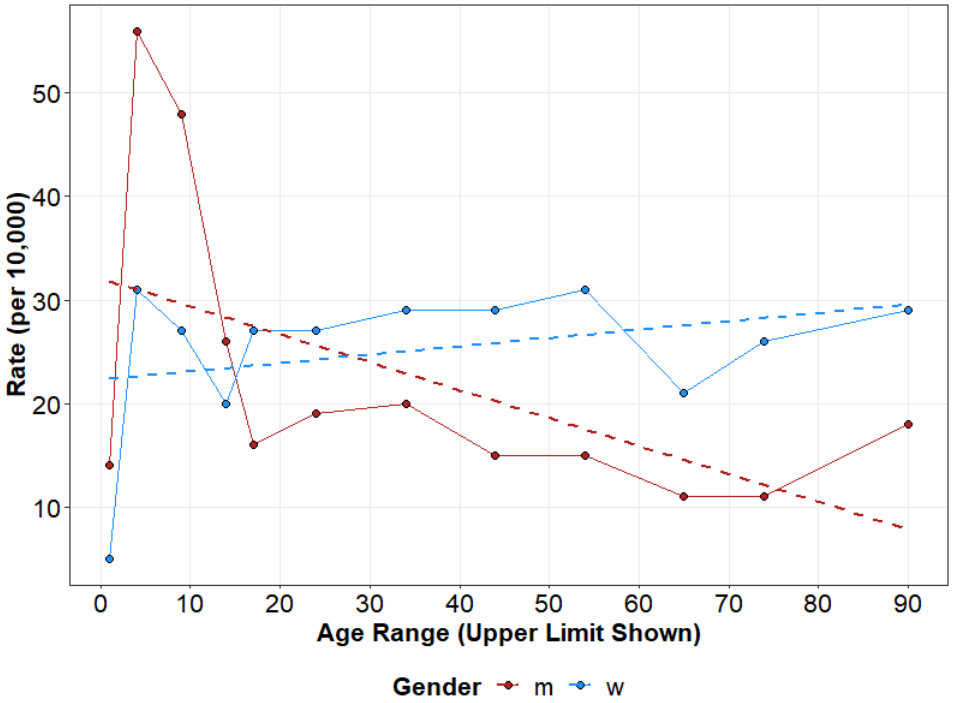

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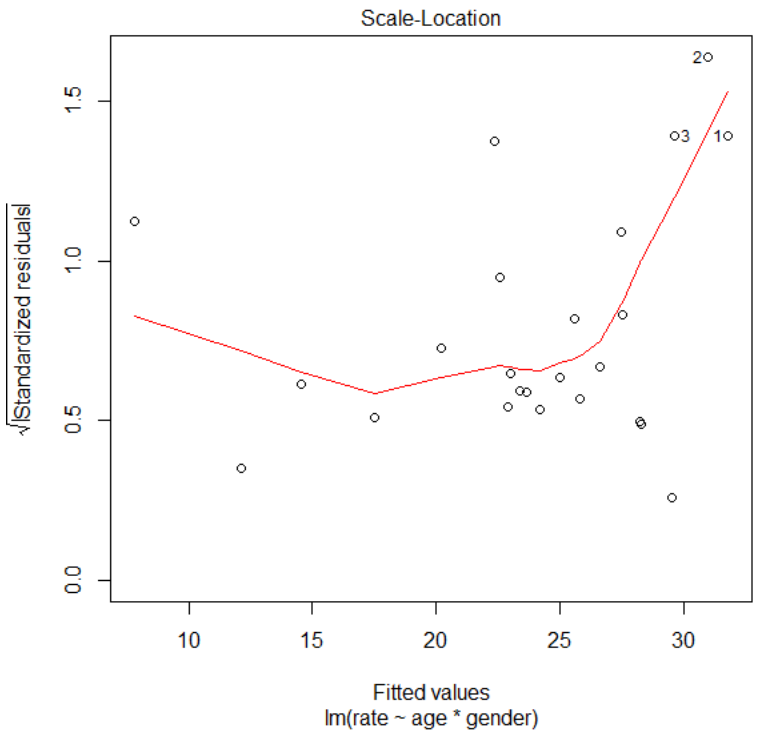

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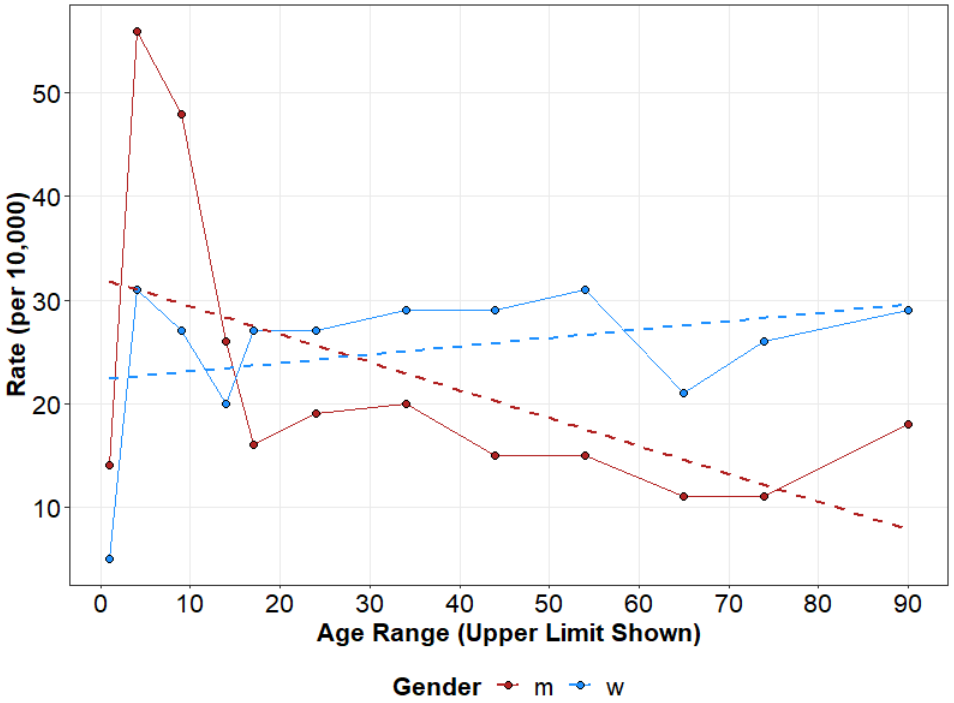

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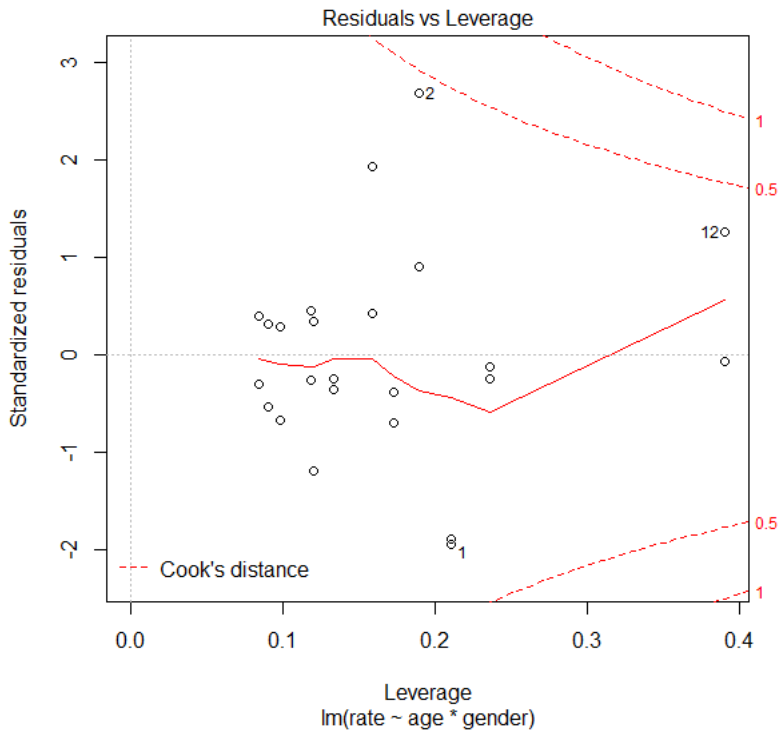

5. Difficult Residuals

- What do we do in these situations?
  - In these situations, we might transform of the outcome variable.
    - E.g., the square-root or log-transformation to remove positive skew.
  - Visually assessing our data, we might also consider different models.
    - E.g., in this example we have a non-linear relationship between Age and Rate, but specifically at low values of Age.
    - Splines, polynomials, or a categorical approach to age might prove to be a better explanation of our data, and lead to better behaved residuals.
  - In cases when homoscedasticity is violated, there are other methods for getting "**robust**" standard errors that lead to more accurate p-values.